

Multiagent Flocking With Angle-Based Formation Shape Control

Gangshan Jing  and Long Wang 

Abstract—This technical note aims to design a decentralized control strategy for multiple double-integrator agents to achieve flocking while maintaining an angle-constrained triangulated formation shape in the plane. We design a novel decentralized formation controller that can steer agents to achieve common velocity while meeting some angle constraints; preserve formation rigidity during agents' motion; and be implemented in the absence of the global coordinate system. Due to the rigidity preservation property, the proposed controller guarantees almost global convergence to a formation satisfying predefined angle constraints. Moreover, we give a definition of “sign” for strongly nondegenerate triangulated frameworks. By implementing the proposed control strategy, the target formation shape is guaranteed to be stable provided its sign is identical to that of the initial formation. A simulation example is presented for illustrating effectiveness of the proposed formation controller.

Index Terms—Flocking, multiagent systems, rigid formation control, rigidity preservation.

I. INTRODUCTION

Decentralized cooperative control of multiagent systems has gained increasing attention from control communities [1]–[5]. Applications can be found in sensor networks, unmanned underwater vehicles, microsatellite clusters, etc. In recent years, different approaches for formation shape stabilization have emerged in the literature [3]–[15]. Among them, a displacement-based formation strategy, which can be treated as a consensus problem [7], [16], usually obtains global convergence of the desired formation shape under a connected formation (sensing) graph [3], [6], [7]. A bearing-based formation strategy can achieve almost global convergence under a rigid formation (sensing) graph [8]. However, both these two approaches require each agent to know the global coordinate system. Another widely studied formation strategy is distance-based formation shape control (also called rigid formation control) [4], [9], [11]–[14], which allows the formation controller to be implemented in the absence of the global coordinate system. Unfortunately, in most of references related to rigid formation control, only local convergence of the target formation shape can be obtained. Moreover, there is no certain condition for initial positions of agents

ensuring that they can eventually form the desired shape. Recently, the authors in [17] presented a novel distance-based approach for achieving almost global convergence of the desired formation shape under a complete formation (sensing) graph. Nevertheless, the sensing graph, especially for networks with large scale, usually cannot be complete in reality. In [18], we proposed an angle-constrained formation controller in the plane, which is also independent of the global coordinate system. Compared to the aforementioned formation approaches, an angle-constrained formation has higher degrees of freedom, thus is more convenient to adjust its scale and orientation. However, similar to distance-based formation, only local convergence of angle errors was obtained in [18].

In this note, by considering the target formation shape as a strongly nondegenerate triangulated framework, we design a decentralized angle-based formation controller to capture almost global convergence of angle errors. Here, “almost global convergence” means that the set of initial configurations such that angle errors do not converge to zero is of measure zero. Different from [18], the formation controller proposed in this note is induced by a novel potential function, and each agent is considered to have double-integrator dynamics. Actually there are many papers concerning flocking control for double-integrator agents (e.g., [20], [21]), the target collective behavior in this note is flocking while maintaining an angle-constrained formation shape. Similar issue in rigid formation control has also been discussed, e.g., [13], [14]. The novel feature of this technical note is that we obtain certain initial conditions for convergence of the desired formation shape.

Our main contributions are summarized as follows.

- 1) Based on the angle rigidity theory, we give a definition of “sign” for triangulated frameworks in the plane. Benefiting from this definition, we derive a necessary and sufficient condition for determining the shape of a triangulated framework in the desired clockwise by angles only (see Lemma 2).
- 2) We propose a decentralized angle-based formation law that can not only drive agents to meet some angle constraints, but also preserve angle rigidity of the formation during agents' motion. Different from the rigidity preservation approach in [22], our method does not require communication-based estimators.
- 3) With the proposed control law, we obtain almost global convergence of the angle errors when the target formation shape is a strongly nondegenerate triangulated framework (called the triangulated formation) (see Theorem 1). In [23], the authors also showed that all stable configurations satisfy the predefined interagent distance constraints, but did not obtain the condition for stabilizing the desired formation shape. In this technical note, we show that the desired formation shape can be certainly asymptotically achieved provided its sign is identical to that of the initial formation (see Theorem 1).

The remainder of this note is organized as follows. Section II introduces some definitions and results associated with the angle rigidity theory. Section III presents the problem that we aim to solve in this note. Section IV presents a solution for the problem of flocking with

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formation shape control. Section V gives a simulation example for illustrating the theoretical result. Section VI concludes this note and shows some possible future work directions. The proof of Lemma 2 is given in the Appendix.

Notations: Throughout this note, $\|\cdot\|$ denotes the Euclidean norm; $\otimes$ is the Kronecker product; $\text{null}(X)$ and $\text{rank}(X)$ denote the null space and the rank of matrix X , respectively; $A \setminus B$ is the set of elements belonging to A but not belonging to B ; $O(2)$ is the orthogonal group in $\mathbb{R}^2$; $SO(2)$ is the special orthogonal group in $\mathbb{R}^2$; $\mathcal{R}_o(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ is the two-dimensional rotation matrix associated with $\theta \in [0, 2\pi)$; and $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ represents an undirected graph, where $\mathcal{V} = \{1, \dots, n\}$ is the vertex set and $\mathcal{E} \subset \mathcal{V}^2$ with $|\mathcal{E}| = m$ is the edge set. Note that we do not distinguish (i, j) and (j, i) in $\mathcal{E}$.

II. PRELIMINARIES OF ANGLE RIGIDITY

In [18], we presented the angle rigidity theory, which is to answer that when a geometric shape in the plane can be determined by subtended angles only. In what follows, we review some definitions and results associated with the angle rigidity theory that will be utilized in this note.

A pair $(\mathcal{G}, p)$ is said to be a framework, where $\mathcal{G}$ is a graph and $p = (p_1^T, \dots, p_n^T)^T$ is called a configuration, and p_i is the coordinate of vertex i , $i = 1, \dots, n$. Let $\mathcal{T}_{\mathcal{G}} = \{(i, j, k) \in \mathcal{V}^3 : (i, j), (i, k) \in \mathcal{E}, j < k\}$, then $\{g_{ij}^T, g_{ik} = A_{ijk}, (i, j, k) \in \mathcal{T}_{\mathcal{G}}\}$ is the set of constraints on all subtended angles in $(\mathcal{G}, p)$, where $A_{ijk} \in [-1, 1]$ and $g_{ij} = \frac{p_i - p_j}{\|p_i - p_j\|}$ is the relative bearing between vertices i and j . The *angle rigidity function* is defined as

$$f_{\mathcal{T}_{\mathcal{G}}}^*(p) = (\dots, g_{ij}^T(p)g_{ik}(p), \dots)^T, \quad (i, j, k) \in \mathcal{T}_{\mathcal{G}}^* \quad (1)$$

where $\mathcal{T}_{\mathcal{G}}^*$ is a subset of $\mathcal{T}_{\mathcal{G}}$ and its elements involve all vertices in $\mathcal{G}$. For the notational simplicity, we adopt $f_{\mathcal{G}}(p) = f_{\mathcal{T}_{\mathcal{G}}}(p)$.

Let $\mathcal{K}$ be the complete graph with n vertices. A framework $(\mathcal{G}, p)$ is *angle rigid* if for some $\mathcal{T}_{\mathcal{G}}^*$, there exists a neighborhood U_p of p such that $f_{\mathcal{T}_{\mathcal{G}}^*}^{-1}(f_{\mathcal{T}_{\mathcal{G}}^*}(p)) \cap U_p = f_{\mathcal{K}}^{-1}(f_{\mathcal{K}}(p)) \cap U_p$. If this equation holds for $U_p = \mathbb{R}^{2n}$, then $(\mathcal{G}, p)$ is *globally angle rigid*. A framework $(\mathcal{G}, p)$ is *minimally angle rigid* if it is angle rigid, and deletion of any edge will make $(\mathcal{G}, p)$ not angle rigid. Let δp be a variation of p , δp is called an *infinitesimal angle motion* corresponding to $\mathcal{T}_{\mathcal{G}}^*$ if $\frac{\partial f_{\mathcal{T}_{\mathcal{G}}^*}}{\partial p} \delta p = 0$. Since a subtended angle is always invariant under rotations, translations, and scalings, these motions are said to be *trivial*. A framework $(\mathcal{G}, p)$ is *infinitesimally angle rigid* if there exists a $\mathcal{T}_{\mathcal{G}}^*$ such that every infinitesimal motion is trivial. In the plane, the trivial motion space is of dimension 4. As a result, $(\mathcal{G}, p)$ is infinitesimally angle rigid if and only if $\dim(\text{null}(R_{\mathcal{T}_{\mathcal{G}}^*}(p))) = 4$, or equivalently, $\text{rank}(R_{\mathcal{T}_{\mathcal{G}}^*}(p)) = 2n - 4$,

where $R_{\mathcal{T}_{\mathcal{G}}^*}(p) \triangleq \frac{\partial f_{\mathcal{T}_{\mathcal{G}}^*}}{\partial p}$ is the *angle rigidity matrix*.

Now, we review a particular class of Laman graphs, called *triangulated Laman graphs* introduced in [23]. A triangulated Laman graph with n vertices ($n \geq 3$) is denoted by $\mathcal{L}_n = (\mathcal{V}_n, \mathcal{E}_n)$, which is obtained as follows. Let $\mathcal{L}_2$ be the graph with vertex set $\mathcal{V}_2 = \{1, 2\}$ and edge set $\mathcal{E}_2 = \{(1, 2)\}$. $\mathcal{L}_n$ is a graph generated by adding a vertex n and two edges (n, i) and (n, j) into graph $\mathcal{L}_{n-1}$ for some i and j satisfying $(i, j) \in \mathcal{E}_{n-1}$. A framework embedded by a triangulated Laman graph is called a *triangulated framework*. A triangulated framework $(\mathcal{L}_n, p)$ is *strongly nondegenerate* if $(p_i^T, p_j^T, p_k^T)^T$ is nondegenerate (i.e., p_i, p_j , and p_k do not stay collinear) for any three vertices i, j, k satisfying $(i, j), (j, k), (i, k) \in \mathcal{E}_n$. The following lemma proposed in [18] will be used in this note.

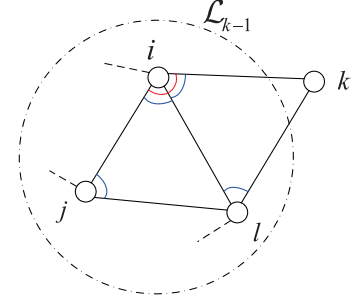


Fig. 1. Assume that $j < l$, the red angle is determined by $(i, j, k) \in \Delta\mathcal{T}_{\mathcal{L}_n}$ and the blue angles are determined by elements in $\mathcal{T}_{\mathcal{L}_n}^*$.

Lemma 1. (see [18]): If $(\mathcal{L}_n, p)$ is strongly nondegenerate, then the following hold.

i) $(\mathcal{L}_n, p)$ is minimally infinitesimally angle rigid for

$$\mathcal{T}_{\mathcal{L}_n}^* = \{(i, j, k) \in \mathcal{V}_n^3 : (i, j), (j, k), (i, k) \in \mathcal{E}_n, i, j < k\}. \quad (2)$$

ii) $(\mathcal{L}_n, p)$ is globally angle rigid for $\mathcal{T}_{\mathcal{L}_n}^* = \mathcal{T}_{\mathcal{L}_n} \cup \Delta\mathcal{T}_{\mathcal{L}_n}$, where $\Delta\mathcal{T}_{\mathcal{L}_n} = \{(i, j, k), j = \min\{\mathcal{N}_i \cap \mathcal{N}_l \cap \mathcal{V}_{k-1}\}, i, l \in \mathcal{N}_k, i < l < k, k = 4, \dots, n\}$ if $n \geq 4$, and $\Delta\mathcal{T}_{\mathcal{L}_n} = \emptyset$ otherwise.

iii) $f_{\mathcal{T}_{\mathcal{L}_n}^*}^{-1}(f_{\mathcal{T}_{\mathcal{L}_n}^*}(p)) = f_{\mathcal{K}_n}^{-1}(f_{\mathcal{K}_n}(p)) = \mathcal{S}_p$, where

$$\mathcal{S}_p = \{q \in \mathbb{R}^{2n} : q = c(I_n \otimes \mathcal{R})p + \mathbf{1}_n \otimes \eta, \mathcal{R} \in O(2), c \in \mathbb{R} \setminus \{0\}, \eta \in \mathbb{R}^2\}. \quad (3)$$

It is worth noting that in a triangulated Laman graph $\mathcal{L}_n$ with $n \geq 4$, for any $k \geq 4$, there are exactly two vertices i and l such that $i, l \in \mathcal{N}_k$ and $i, l < k$, and there is at least one vertex $j \in \mathcal{V}_{k-1}$ being the common neighbor of i and l . Therefore, the set $\Delta\mathcal{T}_{\mathcal{L}_n}$ in Lemma 1 can be uniquely determined, and $|\Delta\mathcal{T}_{\mathcal{L}_n}| = n - 3$. Also note that the index j may be greater or smaller than l . Fig. 1 illustrates an element $(i, j, k) \in \Delta\mathcal{T}_{\mathcal{L}_n}$. Without loss of generality, assume $j < l$, the red angle is determined by $(i, j, k) \in \Delta\mathcal{T}_{\mathcal{L}_n}$, whereas the blue angles are determined by elements in $\mathcal{T}_{\mathcal{L}_n}^*$.

Let $S_{ijk}(p) = \text{sign}(g_{ij}^T(p)\mathcal{R}_o(\frac{\pi}{2})g_{ik}(p))$, we give the following definition.

Definition 1: We define vector

$$S_{\mathcal{L}_n}(p) = (\dots, S_{ijk}(p), \dots)^T, \quad (i, j, k) \in \mathcal{T}_{\mathcal{L}_n}^* \quad (4)$$

as the *sign* of a strongly nondegenerate triangulated framework $(\mathcal{L}_n, p)$.

See Fig. 2 as an illustration of Definition 1.

Based on Lemma 1, we can derive the following lemma.

Lemma 2: Let $(\mathcal{L}_n, p)$ be a strongly nondegenerate framework and $f_{\mathcal{T}_{\mathcal{L}_n}^*}(p) = f_{\mathcal{T}_{\mathcal{L}_n}^*}(q)$. Then

- i) $q \in \mathcal{S}_p$ with $\mathcal{R} \in SO(2)$ if and only if $S_{\mathcal{L}_n}(p) = S_{\mathcal{L}_n}(q)$;
- ii) $q \in \mathcal{S}_p$ with $\mathcal{R} \in O(2) \setminus SO(2)$ if and only if $S_{\mathcal{L}_n}(p) = -S_{\mathcal{L}_n}(q)$.

Proof: See the Appendix. ■

Remark 1: Lemma 2(i) implies that by using the angle constraints $f_{\mathcal{T}_{\mathcal{L}_n}^*}(p) = f_{\mathcal{T}_{\mathcal{L}_n}^*}(q)$ and the sign constraint $S_{\mathcal{L}_n}(p) = S_{\mathcal{L}_n}(q)$, we can not only determine the shape of $(\mathcal{L}_n, p)$ uniquely (i.e., $q \in \mathcal{S}_p$), but also recognize that $(\mathcal{L}_n, p)$ and $(\mathcal{L}_n, q)$ are in the same clockwise or counterclockwise directions (i.e., $\mathcal{R} \in SO(2)$). Similarly, Lemma 2(ii) shows that $(\mathcal{L}_n, p)$ and $(\mathcal{L}_n, q)$ have identical shapes (i.e., $q \in \mathcal{S}_p$), but they are in opposite clockwise or counterclockwise directions (i.e., $\mathcal{R} \in O(2) \setminus SO(2)$).

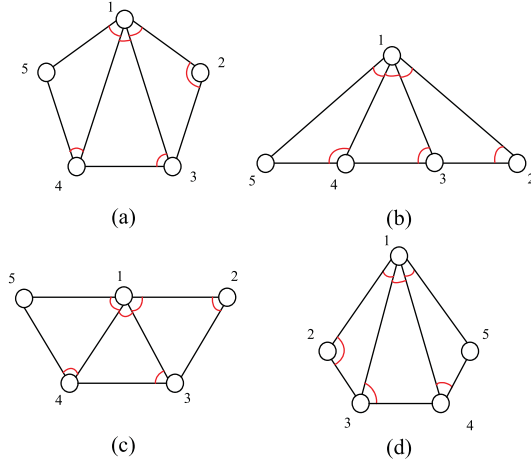


Fig. 2. Four triangulated frameworks. The red angles are angles determined by $\mathcal{T}_{\mathcal{L}_n}^*$. Frameworks (a)–(c) have identical signs and the sign of framework (d) is opposite to that of (a)–(c).

III. PROBLEM STATEMENT

A. Agent Dynamics and Sensing Capability

Consider a group of autonomous agents governed by the following dynamics:

$$\begin{aligned} \dot{p}_i &= v_i \\ \dot{v}_i &= u_i, \quad i \in \mathcal{V} \end{aligned} \quad (5)$$

where p_i and v_i denote the position and velocity, respectively, of agent i , with respect to the global coordinate system and u_i is the control input to be designed. In this note, we use two undirected graphs $\mathcal{G}_p = (\mathcal{V}, \mathcal{E}_p)$ and $\mathcal{G}_v = (\mathcal{V}, \mathcal{E}_v)$ to interpret the position sensing graph and velocity sensing graph, respectively. Moreover, we consider the formation problem in a GPS-denied environment. In this scenario, different agents may have different local coordinate systems, each agent i can only measure $p_i^j - p_j^i$ if $(i, j) \in \mathcal{E}_p$, and $v_i^j - v_j^i$ if $(i, j) \in \mathcal{E}_v$, where p_k^i and v_k^i denote position vector and velocity vector, respectively, of agent k in the local coordinate system of agent i . It is easy to see that $\dot{p}_k^i = v_k^i$.

B. Formation Shape Control

A formation shape control problem is to design a decentralized control law, implementing which the group of agents will be driven to stabilize a desired geometric shape. In this note, we will use angle constraints only to characterize the target formation shape. It is natural to regard the target formation as a framework $(\mathcal{G}, p^*)$, where $p^* = (p_1^{*T}, \dots, p_n^{*T})^T \in \mathbb{R}^{2n}$ is a configuration forming the target formation shape. Similar to the literature [9]–[14], graph $\mathcal{G}$ models not only the formation graph, but also the position sensing graph. That is, $\mathcal{G} = \mathcal{G}_p$. To steer agents to stabilize a formation shape congruent to $(\mathcal{G}, p^*)$ up to translations, rotations, reflections, and scalings, it suffices to drive the stacked vector of positions $p = (p_1^T, \dots, p_n^T)^T$ into $\mathcal{S}_{p^*}$ asymptotically, whose form is shown in (3).

C. Flocking With a Desired Formation Shape

Flocking is a collective behavior describing the phenomenon that all agents move closely with a common velocity. More precisely, usually a flocking behavior is said to be achieved if the group of agents satisfy

three rules [19]: first, velocity consensus; second, cohesion, i.e., the formation scale should be less than a specified value; third, collision avoidance between agents.

In this note, we aim to solve the multiagent flocking problem while steering agents to maintain a desired formation shape. Let $p, v \in \mathbb{R}^{2n}$ be the stacked vectors of positions and velocities, respectively, it is easy to see that the desired equilibrium for (p, v) can be characterized as the following set:

$$\mathcal{E} = \{(p, v) \in \mathbb{R}^{2n} \times \mathbb{R}^{2n} : p \in \mathcal{S}_{p^*}, v \in \text{span}\{\mathbf{1}_n \otimes I_2\}\}. \quad (6)$$

We consider the target formation as a strongly nondegenerate triangulated framework. That is, the following assumption is made.

Assumption 1: $\mathcal{G} = \mathcal{L}_n$, and $(\mathcal{L}_n, p^*)$ is strongly nondegenerate.

Note that if $\mathcal{G}$ contains $\mathcal{L}_n$ as a subgraph and $(\mathcal{L}_n, p^*)$ is strongly nondegenerate, we can still steer agents to stabilize the shape determined by subframework $(\mathcal{L}_n, p^*)$ only and ignore the redundant sensing links, because constraints on $(\mathcal{L}_n, p^*)$ are sufficient to determine the geometric shape of $(\mathcal{G}, p^*)$ uniquely up to translations, rotations, and scalings, as shown in Remark 1.

For the velocity graph, we make the following assumption.

Assumption 2. $\mathcal{G}_v$ is connected.

Let $\mathcal{T}_{\mathcal{G}}^F = \{(i, j, k) \in \mathcal{V}_n^3 : (i, j), (j, k), (i, k) \in \mathcal{E}_n, i, j, k < n\}$, by Lemma 1, $(\mathcal{G}, p^*)$ is infinitesimally angle rigid for $\mathcal{T}_{\mathcal{G}}^F$. We use the following set as the target equilibrium set for the formation system:

$$\begin{aligned} \mathcal{E}_F &= \{(p, v) \in \mathbb{R}^{2n} \times \mathbb{R}^{2n} : f_{\mathcal{T}_{\mathcal{G}}^F}(p) = f_{\mathcal{T}_{\mathcal{G}}^F}(p^*), \\ &\quad v \in \text{span}\{\mathbf{1}_n \otimes I_2\}\}. \end{aligned} \quad (7)$$

Note that $\mathcal{E} \subset \mathcal{E}_F$ and $\mathcal{E} \neq \mathcal{E}_F$. In this note, in dealing with the formation stabilization problem, we will analyze stability of $\mathcal{E}$ and $\mathcal{E}_F$, respectively. Let $\mathcal{N}_i = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$ and $\mathcal{N}_i^v = \{j \in \mathcal{V} : (i, j) \in \mathcal{E}_v\}$, we now formally summarize the problem that will be solved in this note as follows.

Problem 1: (Flocking with formation shape control) Given a target formation $(\mathcal{G}, p^*)$ satisfying Assumption 1, and a velocity graph satisfying Assumption 2, design a distributed control law for each agent i based on the relative position measurements $\{p_i^j - p_j^i, j \in \mathcal{N}_i\}$ and relative velocity measurements $\{v_i^j - v_j^i, j \in \mathcal{N}_i^v\}$, such that $\mathcal{E}$ is asymptotically stable.

IV. FLOCKING WITH FORMATION SHAPE CONTROL

In this section, we aim to design a distributed controller stabilizing $\mathcal{E}_F$. It suffices to minimize the following cost function:

$$\Phi(p, v) = \Phi_p(p) + \Phi_v(v) \quad (8)$$

where $\Phi_p(p) = \frac{1}{2} \sum_{(i,j,k) \in \mathcal{T}_{\mathcal{G}}^F} \delta_{(i,j,k)}^2$ with $\delta_{(i,j,k)} = g_{ij}^T(p)g_{ik}(p) - g_{ij}^T(p^*)g_{ik}(p^*)$ reaches its global minimum when the formation shape is formed and $\Phi_v(p) = \frac{1}{2} \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{N}_i^v} \|v_i - v_j\|^2$ reaches its global minimum when all agents attain a common velocity.

Similar objective functions have been faced in the literature (e.g., [13] and [14]). In their work, for minimizing (8), the control input for each agent is usually designed as $-\nabla_p \Phi_p - \nabla_v \Phi_v$. In this scenario, the desired equilibrium set is $\{\nabla_p \Phi_p = 0 \text{ and } \nabla_v \Phi_v = 0\}$. However, $\delta = 0$ cannot be certainly obtained from $\nabla_p \Phi_p = 0$. To see this, note that $\nabla_p \Phi_p = R_{\mathcal{T}_{\mathcal{G}}^F}^T \delta$. If $R_{\mathcal{T}_{\mathcal{G}}^F}(p)$ is of full row rank, then $\nabla_p \Phi_p = 0$ implies $\delta = 0$, otherwise there always exist undesired equilibria in $\text{null}(R_{\mathcal{T}_{\mathcal{G}}^F}(p))$. Nevertheless, it is difficult to analyze the evolution of $\text{rank}(R_{\mathcal{T}_{\mathcal{G}}^F}(p))$ during the formation process. In [18], $-\nabla_{\mathcal{T}_{\mathcal{G}}^F}^T \delta$ was employed as the control input u , thus there always exist undesired equilibria for the formation system in [18], and it is uncertain whether

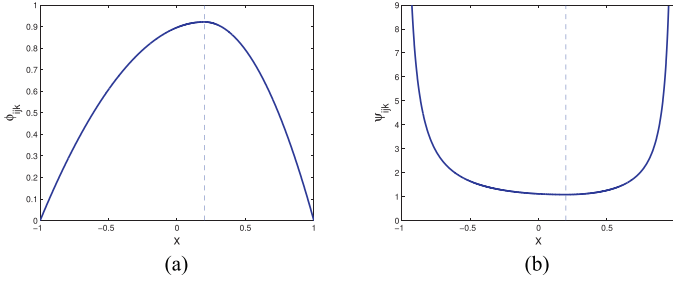


Fig. 3. Artificial potential functions (a) $\varphi_{ijk}(x)$ and (b) $\psi_{ijk}(x)$ for $f_{ijk}^* = 0.2$.

they can be avoided. In this note, we will introduce a control law based on a novel potential function, which is capable of avoiding undesired equilibria, by preserving angle rigidity of the formation during agents' motion.

A. Artificial Potential Function

In this section, we will construct a potential function for inducing a decentralized control law, which can steer agents to meet some angle constraints while maintaining angle rigidity of $(\mathcal{G}, p(t))$ during the evolution process. In [22], the infinitesimal distance rigidity preservation problem is solved by restricting eigenvalues of the rigidity matrix, which requires communication-based estimators. Here, we propose a novel rigidity preservation approach, which can be implemented without interagent communication. The fundamental idea is preserving strong nondegeneracy of $(\mathcal{G}, p(t))$.

By Lemma 1, it suffices to keep p_i, p_j , and p_k not collinear during agents' motion for all $(i, j, k) \in \mathcal{T}_G^F$. We notice that if overlaps do not happen for p_i, p_j , and p_k , then they are collinear if and only if $|f_{ijk}(p)| = |g_{ij}^T(p)g_{ik}(p)| = 1$. As a result, it suffices to keep $f_{ijk} \in (-1, 1)$ for all $t \geq 0$. For notation simplicity, we will adopt $f_{ijk}^* = f_{ijk}(p^*) = g_{ij}^T(p^*)g_{ik}(p^*)$ in the rest of this note.

Let $\psi_{ijk}(\cdot) : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{> 0}$ be a nonnegative function of f_{ijk} , which satisfies the following conditions.

- 1) $\psi_{ijk}(x) \rightarrow \infty$ as $|x| \rightarrow 1$.
- 2) $\psi_{ijk}(x)$ attains its global minimum at $x = f_{ijk}^*$.
- 3) $\psi_{ijk}(x)$ is continuously differentiable everywhere on $(-1, 1)$.

To construct an appropriate function satisfying the above-mentioned three conditions, it suffices to design $\varphi_{ijk} = 1/\psi_{ijk}$ satisfying the following.

- 1) $\varphi_{ijk}(x) = 0$ for $|x| = 1$.
- 2) $\varphi_{ijk}(x)$ attains its global maximum at $x = f_{ijk}^*$.
- 3) $\varphi_{ijk}(x)$ is continuously differentiable everywhere on $[-1, 1]$.

One suitable choice of $\varphi_{ijk}(\cdot)$ is the following piecewise function:

$$\varphi_{ijk}(x) = \begin{cases} -a_{ijk}x^2 + b_{ijk}x + c_{ijk}, & x \in [-1, f_{ijk}^*] \\ -\bar{a}_{ijk}x^2 + \bar{b}_{ijk}x + \bar{c}_{ijk}, & x \in [f_{ijk}^*, 1] \end{cases}. \quad (9)$$

By the three conditions for φ_{ijk} , we have $\varphi_{ijk}(1) = \varphi_{ijk}(-1) = 0$ and $\varphi'_{ijk}(f_{ijk}^*) = 0$, together with continuity of φ_{ijk} , we can derive a set of feasible coefficients: $a_{ijk} = (f_{ijk}^* - 1)^2$, $b_{ijk} = 2a_{ijk}f_{ijk}^*$, $c_{ijk} = (2f_{ijk}^* + 1)a_{ijk}$, $\bar{a}_{ijk} = (f_{ijk}^* + 1)^2$, $\bar{b}_{ijk} = 2\bar{a}_{ijk}f_{ijk}^*$, and $\bar{c}_{ijk} = (1 - 2f_{ijk}^*)\bar{a}_{ijk}$.

In Fig. 3, the two potential functions $\varphi_{ijk}(x)$ and $\psi_{ijk}(x)$ are plotted for $f_{ijk}^* = 0.2$. One can observe that $\psi_{ijk}(x)$ reaches its global minimum at $x = f_{ijk}^*$, and $\psi_{ijk}(x) \rightarrow \infty$ if $|x| \rightarrow 1$. This will ensure formation stabilization and rigidity preservation simultaneously.

Utilizing the potential functions designed earlier, we have the following global potential function for constraining positions of agents:

$$\Phi_F = \sum_{(i,j,k) \in \mathcal{T}_G^F} \psi_{ijk}(f_{ijk}). \quad (10)$$

B. Distributed Formation Stabilization Controller

In solving Problem 1, the agents should not only form a target shape asymptotically, but also achieve a common velocity eventually. To achieve this goal, we hope to minimize the following potential function:

$$\Psi = \Phi_F + \Phi_v \quad (11)$$

where $\Phi_v = \frac{1}{2}v^T \bar{L}v$, $\bar{L} = L \otimes I_2$, and L is the Laplacian matrix of graph $\mathcal{G}_v$.

Now, we propose a formation control law in the following compact form:

$$u = -\nabla_p \Phi_F - \nabla_v \Phi_v = -R_{\mathcal{T}_G^F}^T(p)\xi(p) - \bar{L}v \quad (12)$$

where $u = (u_1^T, \dots, u_n^T)^T$, $\xi = (\dots, \xi_{ijk}, \dots)^T$, and $\xi_{ijk} = \left. \frac{d\psi_{ijk}}{dx} \right|_{x=f_{ijk}}$.

Accordingly, the formation controller for each agent i can be written as

$$u_i = \sum_{(j,k) \in \mathcal{N}_{\mathcal{T}_i}} y_1(p_i - p_j, p_i - p_k) + \sum_{(j,k) \in \mathcal{N}_{\mathcal{T}_i}} y_2(p_j - p_i, p_j - p_k) - \sum_{j \in \mathcal{N}_i^v} (v_i - v_j), i \in \mathcal{V} \quad (13)$$

where $y_1 = -\xi_{ijk}(\frac{\partial f_{ijk}}{\partial p_i})^T$, $y_2 = -\xi_{ijk}(\frac{\partial f_{ijk}}{\partial p_i})^T$, $\mathcal{N}_{\mathcal{T}_i} = \{(j, k) \in \mathcal{V}^2 : (i, j, k) \in \mathcal{T}_G^F\}$, and $\mathcal{N}_{\mathcal{T}_i} = \{(j, k) \in \mathcal{V}^2 : (j, i, k) \text{ or } (j, k, i) \in \mathcal{T}_G^F\}$.

From the definition of $\mathcal{T}_G^F$, for any $(i, j, k) \in \mathcal{T}_G^F$, there must hold $(i, j), (i, k), (k, j) \in \mathcal{E}$. That is, agent i always has access to e_{ij} and e_{ik} , thus it is reasonable for i to obtain e_{jk} by simple subtraction $e_{ik} - e_{ij}$. In conclusion, the controller (13) is distributed. Moreover, it is easy to verify that the control law (13) is independent of the global coordinate system. That is, each agent is allowed to implement (13) based on relative displacements measured from neighbors in its local coordinate frame.

Remark 2: The gradient-based control law (13) can only be implemented under an undirected formation (sensing) graph. In fact, the shape of a directed graph may also be stabilizable. For example, a nondegenerate triangle framework $(\mathcal{L}_3, p)$ can be uniquely determined by two subtended angles. If each agent can adjust the angle subtended by itself via some controller based on local information, then only two agents (without loss of generality, 1 and 2) are required to move for stabilizing the triangle. In this case, agent 3 stays stationary and does not require information from others, implying that the sensing graph can be directed. The synthesis of the above-mentioned controller is yet to be done, and will be a topic of our future work.

Next, we show some properties of the formation system under control law (13).

Let $p^o = \frac{1}{n} \sum_{i \in \mathcal{V}} p_i$ denote the centroid of the formation, $\bar{p} = p - \mathbf{1}_n \otimes p^o$ and $\bar{v} = v - \mathbf{1}_n \otimes \dot{p}^o$. According to (5), when controller (13) is applied, the formation system can be equivalently written as

$$\begin{aligned} \dot{\bar{p}} &= \bar{v} \\ \dot{\bar{v}} &= -R_{\mathcal{T}_G^F}^T(\bar{p})\xi(\bar{p}) - \bar{L}\bar{v} \end{aligned} \quad (14)$$

where $\xi(\bar{p}) = (\dots, \xi_{ijk}(\bar{p}), \dots), (i, j, k) \in \mathcal{T}_G^F$.

Observe that Φ_F is a function of $g = (\dots, g_{ij}^T(p), \dots)^T$, and $g_{ij}(p) = g_{ij}(\bar{p})$. Thus, Φ_F is a function of $\bar{p}$. Now, we denote

$$V_1(\bar{p}, \bar{v}) = \Phi_F(\bar{p}) + \frac{1}{2}\|\bar{v}\|^2, \quad V_2(\bar{p}, \bar{v}) = \bar{p}^T \bar{v} + \frac{1}{2}\bar{p}^T \bar{L} \bar{p} \quad (15)$$

and let λ be the minimum nonzero eigenvalue of L .

Lemma 3: Under control law (13), we have $\dot{V}_1 \leq 0$ and $\dot{V}_1 + \lambda \dot{V}_2 \leq 0$.

Proof: According to (14), we have

$$\dot{V}_1 = (\nabla_{\bar{p}} \Phi_F)^T \bar{v} + \bar{v}^T (-\nabla_{\bar{p}} \Phi_F - \bar{L} \bar{v}) = -\bar{v}^T \bar{L} \bar{v} \leq 0.$$

To prove the second inequality, we should note that $R_{T_G^F} \bar{p} = \frac{\partial f}{\partial g} \text{diag}(\frac{P_{ij}}{\|e_{ij}\|}) \bar{H} \bar{p} = 0$, where $\bar{H} = (H \otimes I_2)$ and $H \in \mathbb{R}^{m \times n}$ is the incidence matrix corresponding to graph $\mathcal{G}$. Then

$$\dot{V}_2 = \|\bar{v}\|^2 + \bar{p}^T (-R_{T_G^F} \xi - \bar{L} \bar{v}) + \bar{p}^T \bar{L} \bar{v} = \|\bar{v}\|^2.$$

From the form of $\bar{v}$, we have $\bar{v}^T (\mathbf{1}_n \otimes I_2) = 0$, then $\bar{v}^T \bar{L} \bar{v} \geq \lambda \|\bar{v}\|^2$. It follows that

$$\dot{V}_1 + \lambda \dot{V}_2 = -\bar{v}^T \bar{L} \bar{v} + \lambda \|\bar{v}\|^2 \leq 0. \quad \blacksquare$$

Let $s = \sqrt{\frac{1}{n} \sum_{i \in \mathcal{V}} \|p_i - p^o\|^2}$ denote the scale of the formation, then we have the following result.

Lemma 4: Under control law (13), the velocity of formation centroid is invariant, i.e., $\dot{p}^o = 0$, the formation scale satisfies $s \leq \sqrt{\frac{1}{n} (\frac{12}{\lambda^2} V_1(0) + \frac{4}{\lambda} V_2(0))}$.

Proof: Because $R_{T_G^F} (\mathbf{1}_n \otimes I_2) = \frac{\partial f}{\partial g} \text{diag}(\frac{P_{ij}}{\|e_{ij}\|}) \bar{H} (\mathbf{1}_n \otimes I_2) = 0$, we have

$$\dot{p}^o = \frac{1}{n} (\mathbf{1}_n \otimes I_2)^T \ddot{p} = \frac{1}{n} (\mathbf{1}_n \otimes I_2)^T (-R_{T_G^F} \xi - \bar{L} \bar{v}) = 0.$$

To show the upper bound of the formation scale s , we first notice that

$$\begin{aligned} V_2 &= \frac{1}{2} \bar{p}^T \bar{L} \bar{p} + \bar{p}^T \bar{v} \geq \frac{1}{2} \bar{p}^T \bar{L} \bar{p} \\ &\quad - \frac{\lambda}{4} \|\bar{p}\|^2 - \frac{1}{\lambda} \|\bar{v}\|^2 \geq \frac{\lambda}{4} \|\bar{p}\|^2 - \frac{1}{\lambda} \|\bar{v}\|^2. \end{aligned}$$

From Lemma 3, $\|\bar{v}\|^2 \leq 2V_1(0)$. Moreover, since $V_1 \geq 0$, we have $\lambda V_2 \leq V_1(0) + \lambda V_2(0)$, implying $V_2 \leq \frac{1}{\lambda} V_1(0) + V_2(0)$.

Therefore

$$\begin{aligned} \|\bar{p}\|^2 &\leq \frac{4}{\lambda} \left(V_2 + \frac{1}{\lambda} \|\bar{v}\|^2 \right) \\ &\leq \frac{4}{\lambda} \left(\frac{1}{\lambda} V_1(0) + V_2(0) + \frac{2}{\lambda} V_1(0) \right) \\ &= \frac{12}{\lambda^2} V_1(0) + \frac{4}{\lambda} V_2(0). \end{aligned} \quad (16)$$

Together with the fact that $s = \sqrt{\frac{1}{n} \|\bar{p}\|^2}$, we complete the proof. $\blacksquare$

From Lemma 4, the upper bound of formation scale is affected by graph $\mathcal{G}_v$. Moreover, the greater λ is, the smaller the upper bound of the formation scale will be. As well known, λ is the algebraic connectivity of graph $\mathcal{G}_v$. Hence, we conclude that a velocity graph with dense edges will benefit for preventing the formation from expanding.

Lemma 5: (Rigidity preservation) Under the control law (13), if $(\mathcal{G}, p(0))$ is strongly nondegenerate, then $(\mathcal{G}, p(t))$ is strongly nondegenerate for all $t \geq 0$.

Proof: From Lemma 3, we know $\Phi_F(\bar{p}(t))$ is bounded for all $t \geq 0$. Suppose that at $t^* \geq 0$, $(\mathcal{G}, p(t^*))$ is not rigid. Then, there must exist $(i, j, k) \in \mathcal{T}_G^F$, such that $p_i(t^*), p_j(t^*)$, and $p_k(t^*)$ are collinear. It follows that $|f_{ijk}(t^*)| = 1$. This implies that $\psi_{ijk} \rightarrow \infty$ as $t \rightarrow t^*$. As a result, $\Phi_F(\bar{p}(t)) \rightarrow \infty$ as $t \rightarrow t^*$, which conflicts with boundedness of V_1 . $\blacksquare$

From the proof of Lemma 5, it is easy to obtain the following lemma.

Lemma 6: Under the control law (13), if $(\mathcal{G}, p(0))$ is strongly nondegenerate, then $S_{\mathcal{G}}(p(t)) = S_{\mathcal{G}}(p(0))$ for any $t \geq 0$.

C. Stability Analysis

Theorem 1: Under Assumptions 1 and 2, by implementing the control law (13), the following statements hold.

- If $(\mathcal{G}, p(0))$ is strongly nondegenerate, then (p, v) asymptotically converges into $\mathcal{E}_F$.
- If $S_{\mathcal{G}}(p(0)) = S_{\mathcal{G}}(p^*)$, then (p, v) asymptotically converges into $\mathcal{E}$ with $\mathcal{R} \in \text{SO}(2)$.
- If $S_{\mathcal{G}}(p(0)) = -S_{\mathcal{G}}(p^*)$, then (p, v) asymptotically converges into $\mathcal{E}$ with $\mathcal{R} \in \text{O}(2) \setminus \text{SO}(2)$.

Proof: (i) It suffices to show stability of $\mathcal{E}_F$ for system (14). From Lemma 3, we have $V_1 \geq 0$ and $\dot{V}_1 = -\bar{v}^T \bar{L} \bar{v} \leq 0$. Then, $\|\bar{v}\|$ is bounded. By virtue of Lemma 4, $\|\bar{p}\|$ is also bounded. Therefore, $\bar{p}$ and $\bar{v}$ must stay in a compact set during the evolution. Invoking LaSalle's invariance principle, $\bar{p}$ and $\bar{v}$ asymptotically converge into the largest invariant set in $\mathcal{M} = \{(\bar{p}, \bar{v}) \in \mathbb{R}^{2n} \times \mathbb{R}^{2n} : \bar{v}^T \bar{L} \bar{v} = 0\}$. In [14], the authors showed that the equilibrium set of (14) is $\bar{\mathcal{M}} = \{(\bar{p}, \bar{v}) \in \mathbb{R}^{2n} \times \mathbb{R}^{2n} : R_{T_G^F}(\bar{p})\xi(\bar{p}) = 0, \bar{L}\bar{v} = 0\}$, it is clear that $\bar{\mathcal{M}}$ is also the largest invariant set of $\mathcal{M}$. Note that $(\mathcal{G}, p(0))$ is strongly nondegenerate. From Lemma 5, $R_{T_G^F}$ is always of full row rank. Thus, $R_{T_G^F}(\bar{p})\xi(\bar{p}) = 0$ is equivalent to $\xi(\bar{p}) = 0$. From Assumption 2, $\bar{v}^T \bar{L} \bar{v} = 0$ implies $\bar{v}_i = \bar{v}_j$ for any $i, j \in \mathcal{V}$, together with $\sum_{i \in \mathcal{V}} \bar{v}_i = 0$, we have $\bar{v} = 0$. Hence, $\bar{\mathcal{M}} = \{(\bar{p}, \bar{v}) \in \mathbb{R}^{2n} \times \mathbb{R}^{2n} : \xi(\bar{p}) = 0, \bar{v} = 0\}$ is asymptotically stable. Also note that $\xi = 0$ implies $\frac{d\psi_{ijk}}{dx}|_{x=f_{ijk}(p)} = 0$ for all $(i, j, k) \in \mathcal{T}_G^F$, which follows that $f_{ijk}(p) = f_{ijk}^*$ for all $(i, j, k) \in \mathcal{T}_G^F$. Together with the fact that $\bar{v} = 0$ implies $v \in \text{span}\{\mathbf{1}_n \otimes I_2\}$, it follows that $(\bar{p}, \bar{v}) \in \bar{\mathcal{M}}$ is equivalent to $(p, v) \in \mathcal{E}_F$. Hence, $\mathcal{E}_F$ is asymptotically stable.

(ii) From Lemma 6, we have $S_{\mathcal{G}}(p(t)) = S_{\mathcal{G}}(p(0)) = S_{\mathcal{G}}(p^*)$ for any $t \geq 0$. Next, we consider (p, v) at the steady state. In Lemma 2, we have shown that $p \in \mathcal{S}_{p^*}$ with $\mathcal{R} \in \text{SO}(2)$ if and only if $S_{\mathcal{G}}(p) = S_{\mathcal{G}}(p^*)$ and $f_{ijk}(p) = f_{ijk}(p^*)$ for all $(i, j, k) \in \mathcal{T}_G^F$. That is, $(p, v) \in \mathcal{E}$ with $\mathcal{R} \in \text{SO}(2)$ if and only if $(p, v) \in \mathcal{E}_F$ and $S_{\mathcal{G}}(p) = S_{\mathcal{G}}(p^*)$. Together with statement (i), we obtain asymptotic convergence of $\mathcal{E}$ with $\mathcal{R} \in \text{SO}(2)$. Similarly, the statement (iii) is also true. $\blacksquare$

Since strongly nondegenerate configurations form a dense subset of $\mathbb{R}^{2n}$ (see [23]), we can conclude that $\mathcal{E}_F$ is almost globally asymptotically stable.

In real application, for achieving different objectives, we may require the formation to change its shape from one to another. From Theorem 1, if all these target formation shapes have identical signs, the formation transformation can be certainly accomplished by controller (13).

D. Analysis for Collision Avoidance

Collision avoidance is an important issue in formation control. In bearing-based formation [8], due to utilization of bearings, an implicit assumption for effectiveness of the control law is collision free between neighboring agents. In this note, the control law (13) also requires computation of bearing information. However, our control law itself can

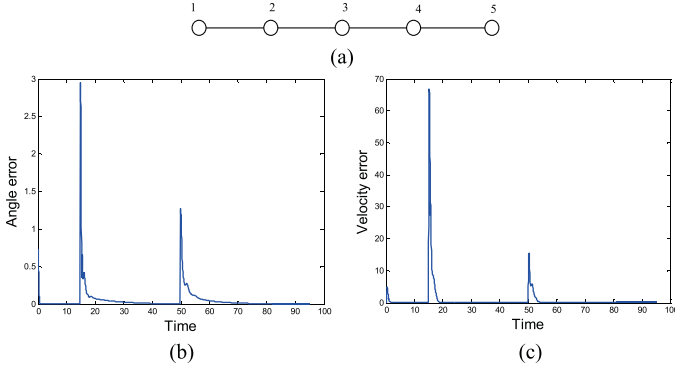


Fig. 4. (a) Velocity graph G_v . (b) Evolution of angle error $\frac{1}{2} \sum_{(i,j,k) \in \mathcal{T}_G^F} (g_{ij}^T(p)g_{ik}(p) - g_{ij}^{*T}(t)g_{ik}^*(t))^2$ and velocity error $\frac{1}{2} \sum_{i \in \mathcal{V}} \sum_{j \in \mathcal{N}_i^v} \|v_i - v_j\|^2$.

ensure collision avoidance between most pairs of neighboring agents in the position graph. See the following lemma.

Lemma 7: (Collision avoidance) Under the control law (13), if $(G, p(0))$ is strongly nondegenerate, then $\|p_i(t) - p_j(t)\| > 0$ for all $t \geq 0$, $(i, j) \in \mathcal{E} \setminus \{(1, 2)\}$.

Proof: Due to definitions of $\mathcal{L}_n$ and $\mathcal{T}_G^F$, for each vertex $k \geq 3$, there must exist $k_1, k_2 \in \mathcal{V}_{k-1}$ such that $(k_1, k_2, k), (k_2, k_1, k) \in \mathcal{T}_G^F$. Note that for any $(i, j) \in \mathcal{E} \setminus \{(1, 2)\}$, at least one of i and j is greater than or equal to 3. Without loss of generality, suppose $i > j \geq 3$. Then, there must hold $j \in \mathcal{V}_{i-1}$ and there exists another vertex $l \in \mathcal{V}_{i-1}$ such that $(l, j, i) \in \mathcal{T}_G^F$. According to the proof of Lemma 5, $f_{ij}(t) = g_{ij}^T(t)g_{il}(t)$ can never be 1 or -1 for $t \geq 0$. If $p_i = p_j$ at some $t^* \geq 0$, then we have $f_{ij}(t^*) = 1$, which is a contradiction. Therefore, $\|p_i(t) - p_j(t)\| > 0$ for all $t \geq 0$. ■

In practice, collision avoidance should be guaranteed between not only adjacent agents, but also nonadjacent agents. In this case, each agent may require to equip a proximity sensor, so that it can sense the relative distance from any agent within the sensing range, and employ a potential-based controller to compute a repulsive force for interagent collision avoidance [20], [21]. Moreover, the collision avoidance between two agents in reality should be in the sense that their distance has a positive lower bound, which can also be achieved by designing the potential function based on proximity sensors.

V. SIMULATION EXAMPLE

In this section, by considering five agents with dynamics (5) moving in the plane, we present an example to illustrate the effectiveness of the formation controller (13). The position graph is identical to the formation graph, the velocity graph is shown in Fig. 4(a).

The objective of this example is to steer the agents to cooperatively achieve flocking while forming the formation shapes shown in Fig. 2(a)–(c) successively. To achieve this goal, the angle constraints $\{g_{ij}^T(p)g_{ik}(p) = g_{ij}^{*T}(t)g_{ik}^*(t), (i, j, k) \in \mathcal{T}_G^F\}$ will be given based on Fig. 2(a)–(c) for $t \in [0, 15]$, $t \in [15, 50]$, and $t \in [50, 95]$, respectively. The potential function (10) can then be induced accordingly. Note that Fig. 2(a)–(c) have identical signs. Let $P \subset \mathbb{R}^{2n}$ denote the set of configurations such that for each $p \in P$, the sign of (G, p) is identical to that of Fig. 2(a)–(c). By setting initial positions of these five agents from P randomly and implementing control law (13), Fig. 4(b) shows the evolution of the angle error and the velocity error and Fig. 5 depicts the trajectories of the agents. One can observe that during each time interval in which the angle constraints remain invariant, the angle error

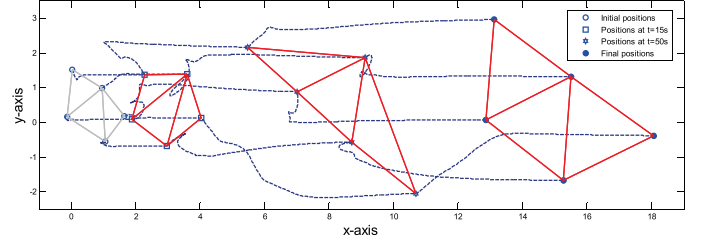


Fig. 5. Trajectories of five agents with the control law (13) in the plane. The formation shapes shown in Fig. 2(a)–(c) are almost perfectly formed at $t = 15$ s, $t = 50$ s, and $t = 95$ s, respectively.

and velocity error converge to zero in a fast speed. Moreover, the target formation shape is almost perfectly formed at $t = 15$ s, $t = 50$ s, and $t = 95$ s, respectively.

VI. CONCLUSION AND FUTURE WORK

We proposed a distributed angle-based control strategy for a group of double-integrator agents to achieve flocking behavior while maintaining a desired triangulated formation shape. Due to the angle rigidity preservation property of the proposed controller, we obtained almost global convergence of angle errors. By defining “sign” of a triangulated framework, we presented a certain initial condition for agents with the control law to achieve the target formation shape.

The angle rigidity theory used in this note is for graphs in two-dimensional space, thus we have only considered formation control in the plane. Our formation controller is still effective if the angle constraints used in the controller are sufficient to determine a geometric shape in three-dimensional (3-D) space. Whether a 3-D shape can be uniquely determined by angles requires an extended angle rigidity theory. A possible approach for angle-based formation control in 3-D space is using a framework consisting of tetrahedrons to describe the target formation.

The future work may also include sign control of a triangulated formation, angle-based formation control for more general target formation shapes, and angle-based formation maneuver control.

APPENDIX PROOF OF LEMMA 2

Let $\mathcal{H}(x) = I_2 - 2xx^T$ be a Householder matrix, where $x \in \mathbb{R}^{2 \times 2}$. The following lemma will be used in proving Lemma 2.

Lemma 8. (see [18]): If $A\eta = B\eta$, where $A, B \in \mathcal{O}(2)$, and $\eta \in \mathbb{R}^2$ is a unit vector, then $A = B$ or $A = B\mathcal{H}_{\eta^\perp}$.

Proof of Lemma 2: In the following proof, for notation simplicity, we denote $U = \mathcal{R}_o(\frac{\pi}{2})$.

(i) Necessity: From $q \in \mathcal{S}_p$, there must hold that $g_{ij}(p) = \mathcal{R}g_{ij}(q)$ for all $i, j \in \mathcal{V}_n$ with $\mathcal{R} \in \mathcal{O}(2)$. Since $\mathcal{R}$ is a rotation matrix, it obviously holds that $g_{ij}^T(p)Ug_{ik}(p) = g_{ij}^T(q)\mathcal{R}^T U g_{ik}(q) = g_{ij}^T(q)Ug_{ik}(q)$ for all $(i, j, k) \in \mathcal{T}_n^*$.

(ii) Sufficiency: We first prove that $f_{\mathcal{K}_n}(p) = f_{\mathcal{K}_n}(q)$. If $n = 3$, the statement obviously holds. Hence, we only consider $n \geq 4$. Due to Lemma 1 (ii), it suffices to show that $g_{ij}^T(p)g_{ik}(p) = g_{ij}^T(q)g_{ik}(q)$ for all $(i, j, k) \in \Delta\mathcal{T}_{\mathcal{L}_n}$.

Given any $(i, j, k) \in \Delta\mathcal{T}_{\mathcal{L}_n}$, let $l = \mathcal{N}_i \cap \mathcal{N}_j \cap \mathcal{N}_k$. Without loss of generality, consider that $i < j < l < k$. Then, $(i, j, l), (j, i, l), (i, l, k), (l, i, k) \in \mathcal{T}_{\mathcal{L}_n}^*$. Since $f_{\mathcal{L}_n}^*(p) = f_{\mathcal{L}_n}^*(q)$, we have $g_{ij}^T(p)g_{il}(p) = g_{ij}^T(q)g_{il}(q)$ and $g_{ji}^T(p)g_{jl}(p) = g_{ji}^T(q)g_{jl}(q)$. Then, $G_{ijl}^T(p)G_{ijl}(p) = G_{ijl}^T(q)G_{ijl}(q)$, where

$G_{ijl}(p) = (g_{ij}(p), g_{il}(p), g_{jl}(p)) \in \mathbb{R}^{2 \times 3}$. Recall that $(\mathcal{L}_n, p)$ is strongly nondegenerate, $\text{rank}(G_{ijl}(p)) = 2$. Hence, there must hold $G_{ijl}(p) = \mathcal{R}_{ijl} G_{ijl}(q)$ for some orthogonal matrix $\mathcal{R}_{ijl} \in \mathbb{R}^{2 \times 2}$. Similarly, we have $G_{ilk}(p) = \mathcal{R}_{ilk} G_{ilk}(q)$. Then, we have $g_{il}(p) = \mathcal{R}_{ijl} g_{il}(q) = \mathcal{R}_{ilk} g_{il}(q)$. According to Lemma 8, $\mathcal{R}_{ijl} = \mathcal{R}_{ilk}$ or $\mathcal{R}_{ijl} = \mathcal{H}(g_{il}^\perp(q)) \mathcal{R}_{ilk}$, where $g_{il}^\perp = \mathcal{R}_o(\frac{\pi}{2}) g_{il}$. Suppose that $\mathcal{R}_{ijl} \neq \mathcal{R}_{ilk}$, then one of them must be a rotation matrix. Without loss of generality, we consider that $\mathcal{R}_{ilk} \in \text{SO}(2)$ and $g_{il}(q) = \mathcal{R}_o(\alpha) g_{ij}$, then $\mathcal{R}_{ijl}$ must be a reflection matrix. Let $U = \mathcal{R}_o(\frac{\pi}{2})$, it follows that $g_{ij}^T(q) U g_{il}(q) = \cos(\alpha + \frac{\pi}{2})$, and $g_{ij}^T(p) U g_{il}(p) = g_{ij}^T(q) \mathcal{R}_{ijl}^T U \mathcal{R}_{ijl} g_{il}(q) = \cos(\alpha - \frac{\pi}{2})$. It is obvious that $\text{sign}(g_{ij}^T(p) U g_{il}(p)) \neq \text{sign}(g_{ij}^T(q) U g_{il}(q))$ and $(i, j, l) \in \mathcal{T}_{\mathcal{L}_n}^*$, a contradiction arises. Hence, we have $\mathcal{R}_{ijl} = \mathcal{R}_{ilk}$, which implies that $g_{ij}^T(p) g_{ik}(p) = g_{ij}^T(q) \mathcal{R}_{ijl}^T \mathcal{R}_{ijl} g_{ik}(q) = g_{ij}^T(q) g_{ik}(q)$.

So far, we have obtained that $f_{\mathcal{K}_n}(p) = f_{\mathcal{K}_n}(q)$. In [18], we showed that there must exist an orthogonal matrix $\mathcal{R} \in \mathbb{R}^{2 \times 2}$ such that $g_{ij}(p) = \mathcal{R} g_{ij}(q)$ for all $i, j \in \mathcal{V}_n$. Suppose $\mathcal{R} \notin \text{SO}(2)$ and let $g_{ik}(q) = \mathcal{R}_o(\alpha) g_{ij}(q)$ for some $(i, j, k) \in \mathcal{T}_{\mathcal{L}_n}^*$. Then, $g_{ij}^T(q) U g_{ik}(q) = \cos(\alpha + \frac{\pi}{2})$ and $g_{ij}^T(p) U g_{ik}(p) = g_{ij}^T(q) \mathcal{R}^T S \mathcal{R} g_{ik}(q) = \cos(\alpha - \frac{\pi}{2})$. This conflicts with $\text{sign}(g_{ij}^T(p) U g_{ik}(p)) = \text{sign}(g_{ij}^T(q) U g_{ik}(q))$. As a result, $\mathcal{R} \in \text{SO}(2)$.

In [8], it has been shown that the shape of an infinitesimally bearing rigid framework can be uniquely determined by bearings up to translations and scalings. Since a strongly nondegenerate triangulated framework is infinitesimally bearing rigid (see [18]), it is straightforward that $q \in \mathcal{S}_p$ if and only if $g_{ij}(p) = \mathcal{R} g_{ij}(q)$ for all $i, j \in \mathcal{V}_n$ with $\mathcal{R} \in \text{SO}(2)$ (note that the sufficiency becomes invalid if infinitesimal bearing rigidity is not satisfied, for example, when the configuration is degenerate).

The statement (ii) can be proved by following similar lines to the aforementioned proof. ■

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